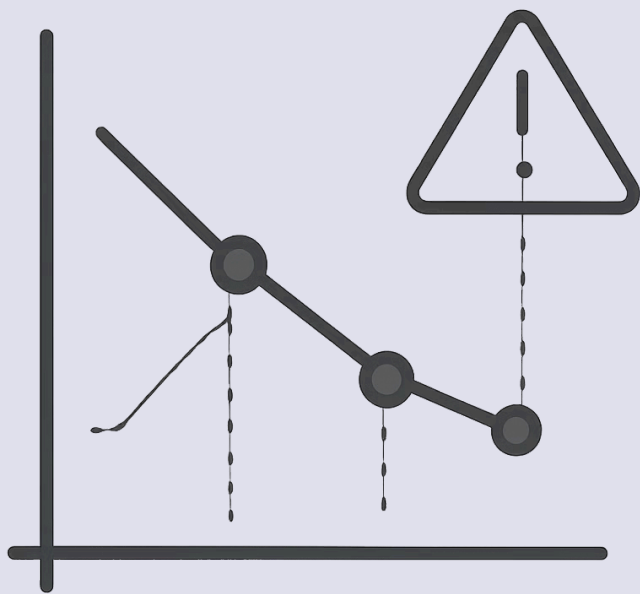




Credit Risk Analysis

Business Objective:

Identify the characteristics of high-risk borrowers and evaluate lending strategies that improve portfolio quality.

255,347 LOANS

11.6% DEFAULT RATE

\$32.6B PORTFOLIO

Portfolio at a Glance

255K

Total Loans

Performing and defaulted combined

11.6%

Default Rate

~1 in 9 borrowers defaults

\$32.6B

Total Exposure

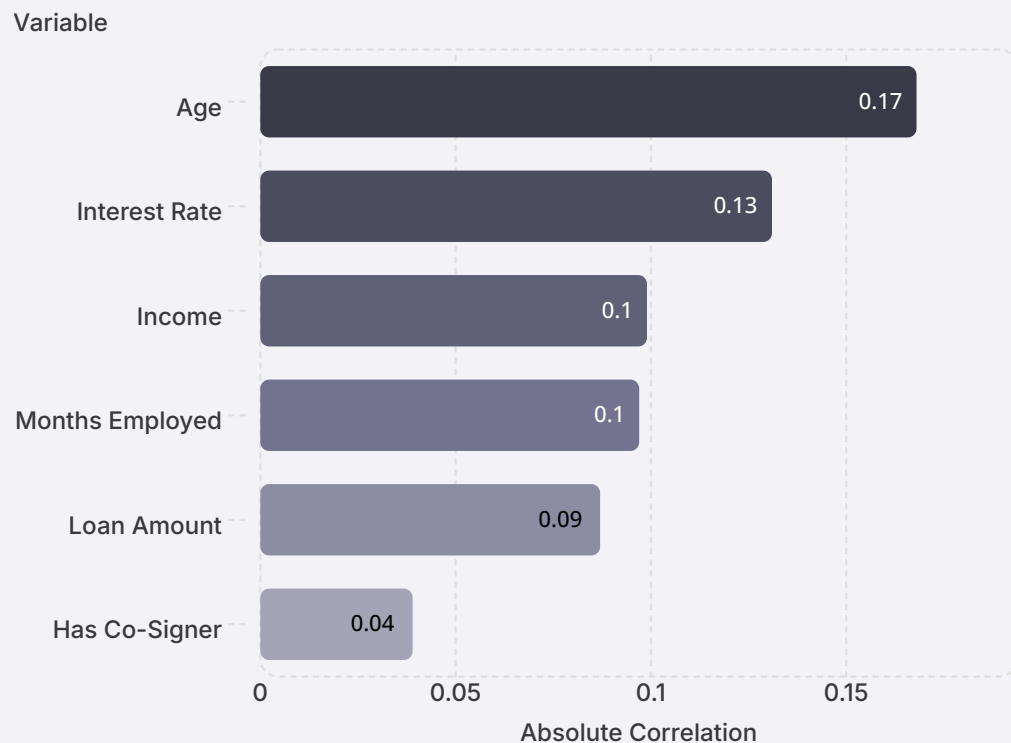
Full portfolio lending volume

\$4.29B

Expected Loss

Projected credit losses across portfolio

Top Risk Drivers: Correlation with Default



What the Data Tells Us

Age and **Interest Rate** are the strongest signals — older borrowers and lower rates correlate with lower default risk.

Income and **employment tenure** show moderate protective effects.

Credit Score has a weak standalone relationship (0.034) — not a dominant predictor in this portfolio.

📌 No single variable strongly predicts default. Risk is multidimensional.

The Defaulting Borrower Profile

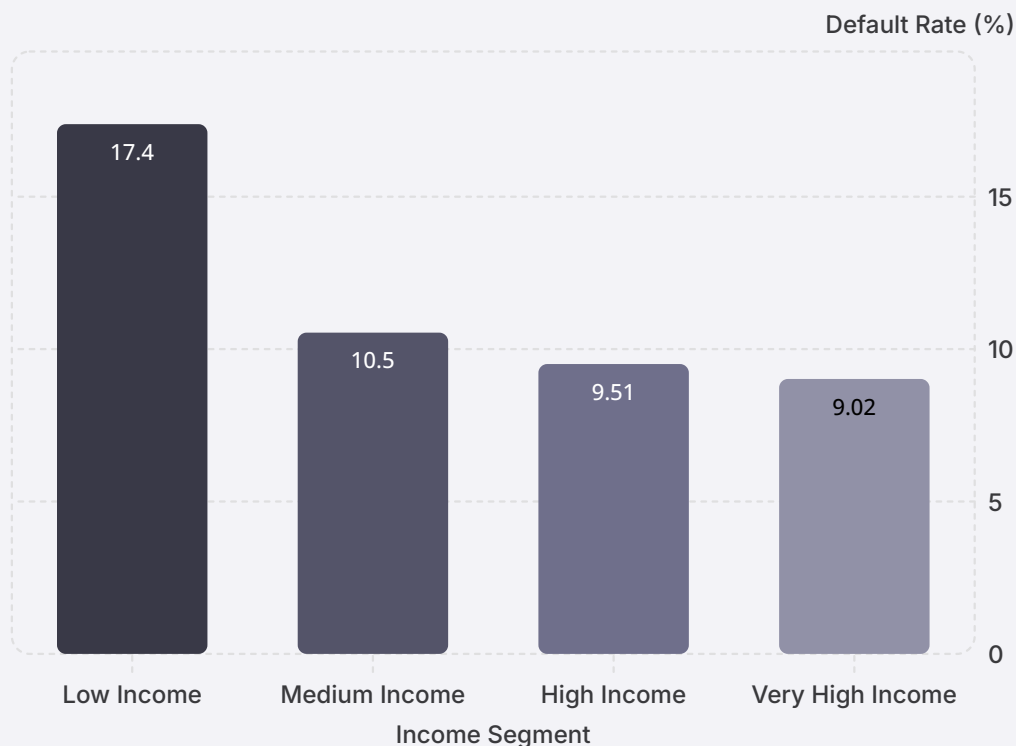
Defaulted borrowers share a consistent financial mismatch: **lower income, larger loans, higher repayment pressure.**



Key Differences

- **Income gap:** Defaulters earn \$12K less on average
- **Loan size:** Defaulters borrow \$19K more on average
- **Credit Score:** Slightly lower (559 vs 576) — not a strong differentiator
- **Employment:** Unemployed borrowers default at 13.6% vs 9.5% for full-time

Income Is the Strongest Risk Segmentation Variable



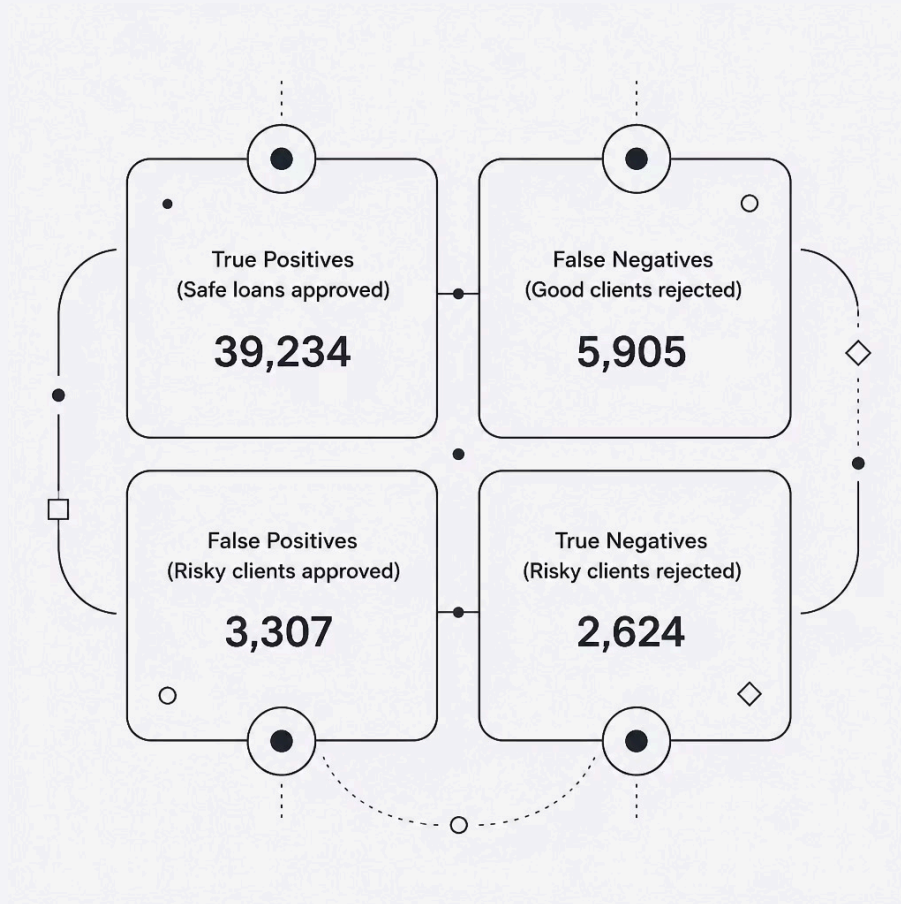
Inverse Relationship: Income vs Default

Low-income borrowers default at **17.4%** — nearly **2× the rate** of high-income borrowers.

Loan amounts are **identical across all segments** (~\$127K), meaning loan size is not risk-adjusted by income.

📄 Low-income segment: 63,837 borrowers · \$8.15B exposure · 17.66% predicted PD

Model Performance at Threshold 0.2



Portfolio Simulation Results

- **Approval rate:** 83% of applications approved
- **Risk in approved portfolio:** 7.8% (down from 11.6% baseline)
- **Net simulated profit:** \$166.4M
- **ROI:** 2.55% on test set exposure
- **ROC-AUC:** 0.751 · **Accuracy:** 88.6%

Model detects only 3% of defaulters — highly conservative in risk flagging, maximizing volume and profit.

Strategy 1: Loan Cap on Low-Income Borrowers



10% Loan Cap — Low-Income Segment

- **Exposure reduction:** \$814.7M
- **Expected loss reduction:** \$173.2M
- **Efficiency:** 21.3 cents saved per \$1 of lending removed
- **Avg loan:** \$127,624 → \$114,862 per borrower

Loan amounts are uniform across income groups — capping low-income borrowers directly reduces concentrated risk without restructuring the portfolio.

Strategy 2: Targeted High-Risk Segment Cap

Ultra-targeted filter: **Low Income + Credit Score < 350 + DTI > 0.60** → 2,109 borrowers, 20.6% observed default rate

10% Cap

Exposure reduced: **\$26.8M**

Loss reduced: **\$6.7M**

Efficiency: **25.1%**

20% Cap

Exposure reduced: **\$53.7M**

Loss reduced: **\$13.5M**

Efficiency: **25.1%**

Combined Hybrid

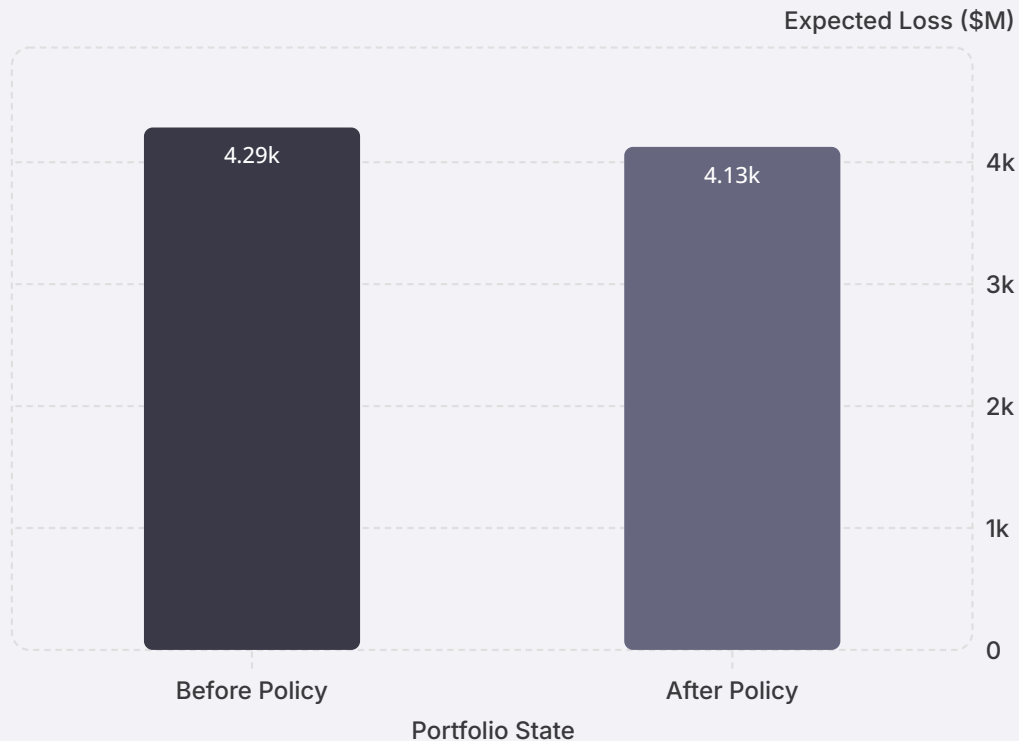
Total exposure reduced: **~\$868M**

Total loss reduced: **~\$179M**

Efficiency: **21-25%**

Strategy 3: High-Risk Borrower Rejection Policy

Reject applications meeting all three criteria: **Credit Score < 400** · **Bottom 25% Income** · **DTI > 55%**



Portfolio Impact

- **5,071 applications** rejected — small fraction of portfolio
- **\$645.6M** lending volume removed
- **\$159.7M** expected loss avoided
- **3.73% reduction** in total expected portfolio loss
- Default rate: 11.61% → 11.44% (-0.17 pp)

Recommended Strategy: Hybrid Approach



Apply 10% Loan Cap — All Low-Income Borrowers

Reduces **\$814.7M** exposure and **\$173.2M** expected loss across 63,837 borrowers

Apply 20% Loan Cap — Ultra High-Risk Segment

Targets Credit Score < 350 + DTI > 0.60 — additional **\$53.7M** exposure and **\$13.5M** loss reduction

Reject: Score < 400 + Low Income + DTI > 55%

Eliminates **\$159.7M** in expected losses by declining only 5,071 highest-risk applications



Combined impact: ~\$868M exposure reduction · ~\$179-\$339M expected loss reduction · 21-25% efficiency